Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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1640 Riverside Drive — Hill Valley, California 11249

Marsha Mallowblack +1 (212) 555 2368 • marshmallow@grad.edu

http://smoresnmore.com • https://github.com/marshmallow

Education

Ph.D. Mathematics, *Spring 2020*
UC Sunnydale; Smallville, Kansas
Thesis Advisor: Dr. Krystal Ball

M.S. Mathematics, *May 2017*
UC Sunnydale; Smallville, Kansas

B.A. Mathematics, *May 2015*
Wossamotta U; Uthaca, NY
Research Interests

CalculusCurves, Integrals, Derivatives, Optimization, Multivariable Calculus, Dot Products, Green's Theorem, Mean Value Theorem, Not So Mean—Actually Kind of Nice—Value Theorem

Linear AlgebraMatrices, Determinants, Vector Spaces

Other InterestsMathematics Education, Mathematical Outreach, Sleeping
Employment
UC Sunnydale Graduate Instructor, Fall 2015 – Present

Wossamotta U Academic Assistant, January 2014 – May 2015
Wossamotta U Mathematics Help Room, October 2013 – May 2015
Teaching Experience
Graduate Instructor
UC Sunnydale MAT 000: Introductory MathematicsFall 2017, Spring 2018, Summer 2019 MAT 555: Intermediate MathematicsSpring 2017, Summer 2017, Fall 2019 MAT 999: Advanced MathematicsSummer 2018, Fall 2018, Spring 2019
Teaching Assistant
UC Sunnydale MAT 000: Introductory MathematicsFall 2015, Spring 2016 MAT 555: Intermediate MathematicsSummer 2016 MAT 999: Advanced MathematicsFall 2016
Wossamotta U Teaching Assistant: General Mathematics IISpring 2015 Teaching Assistant: General Mathematics IFall 2014 Teaching Assistant: General Mathematics IIISpring 2013
Mathematics Clinics & Tutoring
Wossamotta U Academic AssistantJanuary 2014 – May 2015 Mathematics Help RoomOctober 2013 – May 2015
Grants & Awards
Grants

Gov Grant, AAA 9999999 – Funding Program
Awards

Most Awesome CV

Teaching Teaching Assistant Every

Best Imaginary Juggling Routine
Departmental & University Service
Departmental Service
UC Sunnydale [itemsep=0.1cm]

Conference Organizer, *2019*

Identity Theft Specialist, *Spring 2018*

Yelp Reviewer, *Fall 2017*
University Service
UC Sunnydale [itemsep=0.1cm]

Teaching Training, *2016*
Advising
Directed Reading Program Rita Book, Spring 2019Project: How to be Employable with Mathematics
Presentations
Research Talks
[itemsep=0.1cm]

Using Calculus to Intimidate OthersMarch 2020The Biggest Math Conference

Learning to Fear IntegrationSeptember 20191st Annual Mathematical Phobia Conference

How Calculus Earns You FriendshipsApril 20185th Annual Mathematical Conference of Mathematics
University & Department Talks
[itemsep=0.1cm]

How not to get the University suedOctober 2017UC Sunnydale University Seminar
Expository Talks
[itemsep=0.1cm]

Who Are We? Why Are We Here?August 2015UC Sunnydale Math Seminar
Conferences and Workshops
Conferences [itemsep= 0cm]

Annual Mathematics Conference in Graduate MathematicsApril 2019
Workshops [itemsep=0.1cm]

150th Annual Workshop in Bird House BuildingMarch 2019
Skills

Mathematics Programs: Sage, MAGMA, Mathematica, MATLAB, Minitab, SPSS, Geogebra, Surface Evolver, (La)TeX

Presentational Programs: Beamer, Microsoft Word, Microsoft Powerpoint, Microsoft Excel, Pages, Keynote, Inkscape, GoogleDocs, GoogleForms, GoogleSheets, GoogleSlides

Programming: C++, Python, GoogleScript

Languages: English, Spanish (Intermediate), German (Beginner), French (Beginner)

Impressions: Great impression of a hotdog

Music: Recorder

Other Skills: Basket Weaving